Final Practice

Lecture Summary

Please review the course material carefully.

Practice Problems

Question 1: State Taylor's Theorem as given in the lecture. For the function $f(x) = \cos x$ with $x_0 = 0$, compute the Taylor polynomial $P_3(x)$ and the remainder $R_3(x)$. Then, find a bound on $|R_3(x)|$ for $x \in [0, \pi/2]$.

Question 2: Describe the floating-point representation in the 6-bit precision system from the lecture, including the cases for subnormals, infinity, and NaN. Convert the binary representation 1 011 11 to decimal using this system, showing all steps.

Question 3: For the function $f(x) = x^3 - x - 2$, verify using the Intermediate Value Theorem (IVT) that there is a root in $[1, 2]$. Then, apply the bisection method starting with $a = 1, b = 2$, and compute the first three iterations, showing the updated intervals and midpoints. Discuss the convergence rate.

Question 4: Consider the function $f(x) = x^3 - 2x + 1$ on the interval $[0, 1]$. Starting with initial guesses $p_0 = 0.5$ and $p_1 = 0.6$, apply the Secant Method (Algorithm 2.4) for 3 iterations to approximate the root, and compute the absolute error at each step.

Question 5: Note that $f(x) = (x-2)^3(x+1)$ has a root at $p = 2$ of multiplicity $m = 3$. Apply the modified Newton's method iteration $p_{n+1} = p_n - \frac{f(p_n)f'(p_n)}{[f'(p_n)]^2 - f(p_n)f''(p_n)}$ starting from $p_0 = 2.1$ for 2 iterations. Compare the convergence speed to standard Newton's method.

Question 6: Use Steffensen's Method to accelerate the fixed-point iteration for solving $x = \cos(x)$ with initial guess $p_0 = 0.7$. Perform one full loop of the algorithm (computing p_1, p_2, and the accelerated p) and verify if it approximates the fixed point (approximately 0.739085) better than standard fixed-point iteration. Assume tolerance $\epsilon = 0.01$ and discuss quadratic convergence per Theorem 2.15.

Question 7: Prove that polynomial interpolation of degree at most n is unique for $n + 1$ distinct points.

Question 8: Given points $x_0 = 0, f_0 = 1$; $x_1 = 1, f_1 = 2$; $x_2 = 3, f_2 = 10$, write the Newton interpolating polynomial $P_2(x)$ in nested form.

Question 9: Using the forward difference table for $f_j = j^2$ at $j = 0, 1, 2, 3$: $f_0 = 0, f_1 = 1, f_2 = 4, f_3 = 9$, compute $\Delta^2 f_1$ and the general pattern for constant second differences.

Question 10: Derive the system of equations for the coefficients c_j (for $j = 0, 1, 2$) to approximate $f'(x_i)$ using a second-order finite difference scheme at three equidistant points x_{i-1}, x_i, x_{i+1} with spacing h, assuming the approximation is $f'(x_i) \approx c_{i-1}f(x_{i-1}) + c_i f(x_i) + c_{i+1}f(x_{i+1})$.

Question 11: Derive the error term for the trapezoidal rule approximating $\int_a^b f(x)\,dx$ with $h = b - a$.

Question 12: For the ODE $y' = y$, $y(0) = 1$, with exact solution $y(t) = e^t$, apply forward Euler with step $h = 0.5$ to find y_1 and bound the error at $t = 0.5$ using Theorem 5.9 (assume $L = 1$, $M = e^{0.5}$).

Question 13: Write the Butcher tableau for the classical fourth-order Runge-Kutta method and explain how to compute one step for the ODE $y' = f(t, y)$ from y_i to y_{i+1} with step size h.

Solutions

Question 1: Taylor's Theorem: If $f \in C^n([a, b])$ and $f^{(n+1)}$ exists on (a, b), then

$$f(x) = P_n(x) + R_n(x),$$

where

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k,$$

and

$$R_n(x) = \frac{f^{(n+1)}(\gamma(x))}{(n+1)!}(x - x_0)^{n+1},$$

with $\gamma(x)$ between x and x_0. For $f(x) = \cos x$, derivatives: $f'(x) = -\sin x$, $f''(x) = -\cos x$, $f'''(x) = \sin x$, $f^{(4)}(x) = \cos x$. At $x_0 = 0$: $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$. Thus,

$$P_3(x) = 1 + 0 \cdot x + \frac{-1}{2}x^2 + \frac{0}{6}x^3 = 1 - \frac{x^2}{2}.$$

Remainder:

$$R_3(x) = \frac{f^{(4)}(\gamma(x))}{4!}x^4 = \frac{\cos(\gamma(x))}{24}x^4,$$

where $\gamma(x)$ is between 0 and x. For $x \in [0, \pi/2]$, $|\cos(\gamma(x))| \leq 1$, so

$$|R_3(x)| \leq \frac{1}{24}x^4 \leq \frac{1}{24}\left(\frac{\pi}{2}\right)^4.$$

Question 2: The 6-bit system: 1 sign bit (s), 3 exponent bits (c, bias $c_0 = 3$), 2 fraction bits (f). General form:

$$x = \begin{cases} (-1)^s \cdot 2^{c-3} \cdot (1 + f) & \text{if } 1 \leq c \leq 6, \\ (-1)^s \cdot 2^{-2} \cdot f & \text{if } c = 0 \text{ (\textit{subnormal})}, \\ \pm\infty & \text{if } c = 7, f = 0, \\ \text{NaN} & \text{if } c = 7, f \neq 0. \end{cases}$$

Here, $f = f_1 \cdot 2^{-1} + f_2 \cdot 2^{-2}$. For 1 011 11: $s = 1$ (negative), $c = 011$ binary $= 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1 = 3$, $f = 11$ binary $= 1 \cdot 2^{-1} + 1 \cdot 2^{-2} = 0.5 + 0.25 = 0.75$.

Since $c = 3$ (between 1 and 6),

$$x = (-1)^1 \cdot 2^{3-3} \cdot (1 + 0.75) = -1 \cdot 1 \cdot 1.75 = -1.75.$$

Question 3: IVT application: f is continuous (polynomial). $f(1) = 1 - 1 - 2 = -2 < 0$, $f(2) = 8 - 2 - 2 = 4 > 0$, opposite signs, so $\exists p \in (1, 2)$ with $f(p) = 0$.

Bisection iterations:

Iteration 1: $p_1 = (1 + 2)/2 = 1.5$, $f(1.5) \approx 3.375 - 1.5 - 2 = -0.125 < 0$. Same sign as $f(1) < 0$, so root in $(1.5, 2)$, set $a_2 = 1.5, b_2 = 2$.

Iteration 2: $p_2 = (1.5 + 2)/2 = 1.75, f(1.75) \approx 5.359 - 1.75 - 2 \approx 1.609 > 0$. Same sign as $f(2) > 0$, so root in $(1.5, 1.75)$, set $a_3 = 1.5, b_3 = 1.75$.

Iteration 3: $p_3 = (1.5 + 1.75)/2 = 1.625, f(1.625) \approx 4.291 - 1.625 - 2 \approx 0.666 > 0$. Same sign as $f(1.75) > 0$, so root in $(1.5, 1.625)$, set $a_4 = 1.5, b_4 = 1.625$.

Convergence: Interval halves each time, error bound $|p_n - p| \leq (b-a)/2^n$, linear convergence with rate $1/2$.

Question 4: Secant Method iterations (Algorithm 2.4): Initial: $p_0 = 0.5$, $q_0 = f(0.5) = 0.125$; $p_1 = 0.6$, $q_1 = f(0.6) \approx 0.016$. Iteration 1: $p_2 = 0.6 - \frac{0.016 \cdot (0.6-0.5)}{0.016-0.125} \approx 0.6147$, absolute error to $p \approx 0.6180$ is ≈ 0.0033. Iteration 2: $p_3 \approx 0.6179$, error ≈ 0.0001. Iteration 3: $p_4 \approx 0.6180$, error ≈ 0.0000 (converged within machine precision).

Question 5: Modified Newton's Method iterations starting from $p_0 = 2.1$: Iteration 1: $p_1 \approx 1.9990$. Iteration 2: $p_2 \approx 2.0000$.

Comparison to standard Newton's Method: Standard: Iteration 1 ≈ 2.0670, Iteration 2 ≈ 2.0448.

Question 6: Fixed-point iteration for $x = \cos(x)$, $g(x) = \cos(x)$, initial $p_0 = 0.7$.

One full loop of Steffensen's Method: $p_1 = \cos(0.7) \approx 0.7648$. $p_2 = \cos(p_1) \approx 0.7215$. Accelerated $p = p_0 - (p_1 - p_0)^2/(p_2 - 2p_1 + p_0) \approx 0.7389$.

Verification: True fixed point ≈ 0.739085. Steffensen's $p \approx 0.7389$ (error ≈ 0.0002), closer than standard $p_2 \approx 0.7215$ (error ≈ 0.0176). Per Theorem 2.15, converges quadratically near the root ($g'(p) \approx -\sin(0.739) \approx -0.675 \neq 0$ or 1, and g'' bounded).

Question 7: Suppose $P(x)$ and $Q(x)$ both interpolate $f(x)$ at $x_0, \ldots, x_n$, i.e., $P(x_k) = Q(x_k) = f(x_k)$ for $k = 0, \ldots, n$. Let $D(x) = P(x) - Q(x)$, degree $\leq n$, with $D(x_k) = 0$ for $n+1$ points. A non-zero polynomial of degree $\leq n$ has at most n roots unless identically zero. Thus $D(x) \equiv 0$, so $P = Q$.

Question 8: Divided differences: $f[x_0] = 1$, $f[x_1] = 2$, $f[x_2] = 10$ $f[x_0, x_1] = \frac{2-1}{1-0} = 1$ $f[x_1, x_2] = \frac{10-2}{3-1} = 4$ $f[x_0, x_1, x_2] = \frac{4-1}{3-0} = 1$ Newton form: $P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) = 1 + 1 \cdot x + 1 \cdot x(x-1) = 1 + x + x^2 - x = x^2 + 1$.

Question 9: Differences:

$\Delta f_0 = 1 - 0 = 1$, $\Delta f_1 = 4 - 1 = 3$, $\Delta f_2 = 9 - 4 = 5$, $\Delta^2 f_0 = 3 - 1 = 2$, $\Delta^2 f_1 = 5 - 3 = 2$

Second differences are constant 2, as for quadratic, higher differences zero. General: For polynomial degree k, $\Delta^{k+1} = 0$.

Question 10: To derive the system, use Taylor expansions around x_i:

$f(x_{i-1}) = f(x_i) - hf'(x_i) + \frac{h^2}{2}f''(x_i) - \frac{h^3}{6}f'''(\xi_1)$, $f(x_i) = f(x_i)$, $f(x_{i+1}) = f(x_i) + hf'(x_i) + \frac{h^2}{2}f''(x_i) + \frac{h^3}{6}f'''(\xi_2)$.

Substitute into the approximation and ignore $O(h^3)$ terms for second-order accuracy:

$$f'(x_i) \approx c_{i-1}\left(f(x_i) - hf'(x_i) + \frac{h^2}{2}f''(x_i)\right) + c_i f(x_i) + c_{i+1}\left(f(x_i) + hf'(x_i) + \frac{h^2}{2}f''(x_i)\right)$$

Collect terms:

Constant in $f(x_i)$: $c_{i-1} + c_i + c_{i+1} = 0$. Coefficient of $f'(x_i)$: $-hc_{i-1} + hc_{i+1} = 1$. Coefficient of $f''(x_i)$: $\frac{h^2}{2}c_{i-1} + \frac{h^2}{2}c_{i+1} = 0$

Simplifying, the system is:

$$\begin{cases} c_{i-1} + c_i + c_{i+1} = 0 \\ -c_{i-1} + c_{i+1} = \frac{1}{h} \\ c_{i-1} + c_{i+1} = 0 \end{cases}$$

Solving: From the third equation, $c_{i+1} = -c_{i-1}$. Plug into second: $-c_{i-1} - c_{i-1} = \frac{1}{h} \implies c_{i-1} = -\frac{1}{2h}, c_{i+1} = \frac{1}{2h}$. From first: $c_i = 0$.

Question 11: Trapezoidal: $\int_a^b f(x)\,dx \approx \frac{h}{2}(f(a) + f(b))$. Error $E = \int_a^b (f(x) - P_1(x))\,dx$, where $P_1(x)$ is linear interpolant. From interpolation error, $f(x) - P_1(x) = \frac{f''(\xi(x))}{2}(x-a)(x-b)$. Thus $E = \int_a^b \frac{f''(\xi(x))}{2}(x-a)(x-b)\,dx$. By mean value theorem for integrals, $E = \frac{f''(\eta)}{2}\int_a^b (x-a)(x-b)\,dx$ for some $\eta \in (a,b)$. Compute integral: Let $u = x - a$, then $\int_0^h u(u-h)\,du = \int_0^h (u^2 - hu)\,du = \left[\frac{u^3}{3} - \frac{hu^2}{2}\right]_0^h = \frac{h^3}{3} - \frac{h^3}{2} = -\frac{h^3}{6}$. So $E = \frac{f''(\eta)}{2}\left(-\frac{h^3}{6}\right) = -\frac{h^3}{12}f''(\eta)$.

Question 12: Forward Euler: $y_0 = 1$, $y_1 = y_0 + hf(0, y_0) = 1 + 0.5 \cdot 1 = 1.5$. Error bound: $|y(t_i) - y_i| \leq \frac{hM}{2L}(e^{L(t_i - a)} - 1)$. Here $t_1 = 0.5$, $a = 0$, $L = 1$, $M = \sup|y''(t)| = e^{0.5} \approx 1.648$. So bound $= \frac{0.5 \cdot 1.648}{2 \cdot 1}(e^{1 \cdot 0.5} - 1) \approx \frac{0.824}{2}(1.648 - 1) \approx 0.412 \cdot 0.648 \approx 0.267$. Actual error: $e^{0.5} - 1.5 \approx 1.648 - 1.5 = 0.148$, which is less than bound.

Question 13: The Butcher tableau for the classical fourth-order Runge-Kutta is:

0	0	0	0	0
1/2	1/2	0	0	0
1/2	0	1/2	0	0
1	0	0	1	0
	1/6	1/3	1/3	1/6

To compute one step:

$k_1 = f(t_i, y_i)$, $k_2 = f(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_1)$, $k_3 = f(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_2)$, $k_4 = f(t_i + h, y_i + hk_3)$, $y_{i+1} = y_i + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$.

This is derived by matching Taylor series expansions up to order 4 for accuracy.